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Employment of Geochemical Fingerprinting Technology for Dynamic Production Monitoring in Unconventional Wells

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Abstract

Traditional geochemical methods such as Pyrolysis and Soxhlet extraction techniques have been used for decades to guide development programs of conventional reservoirs. However, these conventional practices cannot be extended to meet the required demands of unconventional reservoirs due to the limitations in predicting fracture growth and dynamic production monitoring in unconventional wells. This study presents an integrated workflow to optimize field development using results of time-lapse geochemical fingerprinting technology in unconventional fields.

The methodology involved selecting representative rock samples from tight reservoirs and time-lapse oil samples. Then, high-resolution geochemical fingerprints were acquired from the produced oil samples and the oil samples extracted from the source rocks. By deploying this methodology, a mathematical regression model was built by data mining the resulting fingerprint data based on over 2000 geochemical compounds. Utilizing this model, a group of Reservoir Characterization Indices (RCI) were developed and used as natural tracers to track back the contributing zones of the produced oil and to predict the drainage fracture height to maximize recovery from stacked and staggered reservoir developments.

An integrated workflow has been employed to optimize field developments using geochemical fingerprinting technology in unconventional wells. Integrating the dynamic reservoir monitoring data with the geochemical RCI data has greatly assisted to identify key factors driving the performance of unconventional wells and inter-well interference. The high-resolution analysis of both produced and extracted oil from the cores resulted in dynamically capturing the change in drainage height of the stimulated rock volume over the sampling period. Thereby developing a map of the change in production drainage height in the stimulated reservoir as a function of time. Accurate characterization of the drainage fracture height enabled the determination of the optimum landing point for stacked and staggered placements of unconventional wells. This in turn assisted in improving overall hydrocarbon recovery and economics from shale reservoirs.

This study presents a pioneering workflow to characterize the dynamic drainage fracture height and zonal relative contribution by measuring the geochemical fingerprints of source rocks and produced oil samples. The employment of this unique approach has significantly assisted to determine the optimal well placement and to improve long-term development strategies of unconventional fields.

Introduction

Every unconventional reservoir comes with unique challenges during initial exploration, field delineation and into full field development. Much information is used to identify key geologic drivers for production and controls on Stimulated Rock Volume (SRV) growth during hydraulic fracturing. An important piece of information that has previously been unattainable in unconventional reservoirs is the oil Drainage Rock Volume (DRV) of a producing well. A new methodology of high-resolution geochemical fingerprinting has been developed which allows for pinpointing where produced oils are draining from in the stratigraphic column at $\sim 5 - 10'$ vertical resolution. While the SRV is created during a one-time event upon hydraulic fracturing, the DRV is dynamic and often changes through time. Monitoring producing wells in early production often gives the best understanding of total SRV contributing to production and this data can be used to constrain frac models (Albrecht et al. 2022, Bachleda et al. 2022, Bachleda et al. 2024, Ge et al. 2022, Ge et al. 2024, Jin et al. 2022, Oyeboode et al 2023). Monitoring producing wells through time also provides key learning opportunities such as what portion of the reservoir contributes the most to long-term sustained production, amount of inter-well communication, as well as how landing zone and rock properties may impact zonal contributions.

One example is provided that utilizes this high resolution multi-dimensional gas chromatography (GCxGC) technology. Results from the first ~ 5 -weeks of production of a 4-well pad that target 2 separate landing zones are reviewed. This project aims to understand the Drainage Frac Height (DFH) of each well at $5'$ vertical resolution, primary contributing zones to production, and how the drainage evolves over early production. Other key aspects of this project will be to understand how reservoir properties impact zonal contribution and the amount of inter-well communication to guide future stacking and spacing designs.

Methodology

Production allocation has been a tool that geochemists have used for decades to understand what stratigraphic interval produced oil is coming from. The technology was originally designed for conventional reservoirs, oil samples from each conventional zone were collected and used as end-members to build a production allocation model. This allowed for quantification of zonal contributions for a well producing from some or all of the end-members. As industry became more active in unconventional plays this approach did not provide sufficient resolution for tight reservoirs, also, collecting pure oil endmembers over low permeability rock is not cost effective and extremely challenging. In an effort to monitor oil contribution from low permeability reservoirs, many have explored additional applications of geochemical fingerprinting technology (Rasdi et al. 2012, Jweda et al. 2017, Liu et al. 2017, Liu et al. 2019, Wright et al. 2019, Liu et al. 2020, Bachleda et al. 2022, Ge et al. 2022, Jin et al. 2022).

This methodology focuses on maximizing the data available to build the production allocation model through use of multidimensional gas chromatography (GCxGC) as well as a patented extraction technique to extract produced oil from each rock (i.e. cuttings or core) endmember. This methodology is able to resolve thousands of chemical compounds which are used to build compound ratios (i.e., geochemical fingerprints) and distinguish producing intervals at up to $\sim 5 - 10'$ vertical resolution in low permeability reservoirs. This approach extracts oil from the reservoir itself to use as endmembers for building the model, and in doing so it answers the question of where the oil is located in present time, not where it was sourced from over geological time. A detailed history and explanation of these methodologies are documented in various literatures (Liu et al. 2017, Liu et al. 2019, Liu et al. 2020, Bachleda et al. 2022, Ge et al. 2022, Jin et al. 2022). Production allocation results can be visualized in many ways, Figure 1 demonstrates 3 visualizations shown in this paper (Ward et al 2024).

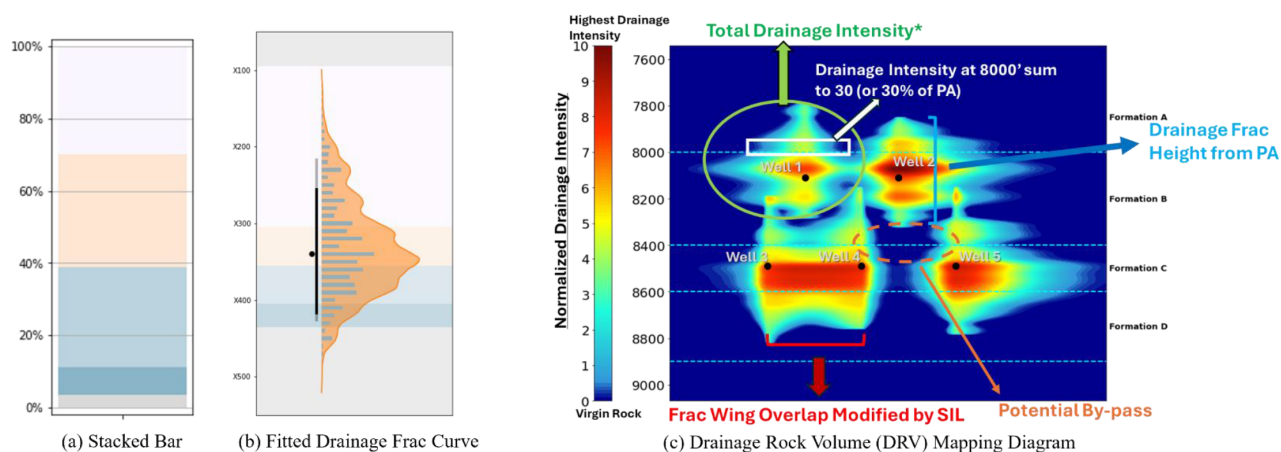


Figure 1—Three Types of Production Allocation visualizations shown; a) Stacked plot showing contribution by formation, b) Fitted Drainage Frac Curve (FDFC) and its P80 and P90 Drainage Frac Height shown by the vertical black and grey bar on the left (representing continuous drainage frac height that covers 80% and 90% of production respectively) and c) DRV mapping diagram to show how Production Allocation and Similarity Index are used to constrain the normalized drainage intensity in the vertical and horizontal directions.

The physical extraction process of oil from the rock also provides important information regarding the reservoir itself and this information is termed Reservoir Characterization Indices (RCI). Two of the most robust RCI measurements are Oil-in-Place Index (OIPI) and Reservoir Quality Index (RQI) shown in Figure 2a. OIPI is often used as a mobile oil saturation indicator and higher OIPI values reflect where pore spaces are filled with more oil than water (Jin et al. 2022). OIPI has been used to better target producing intervals and create more accurate OOIP maps (Bachleda et al. 2022, Bachleda et al. 2023). RQI is a matrix permeability indicator and often corresponds well with NMR T2 distribution, indicating larger pore bodies able to extrude hydrocarbons (Jin et al. 2022).

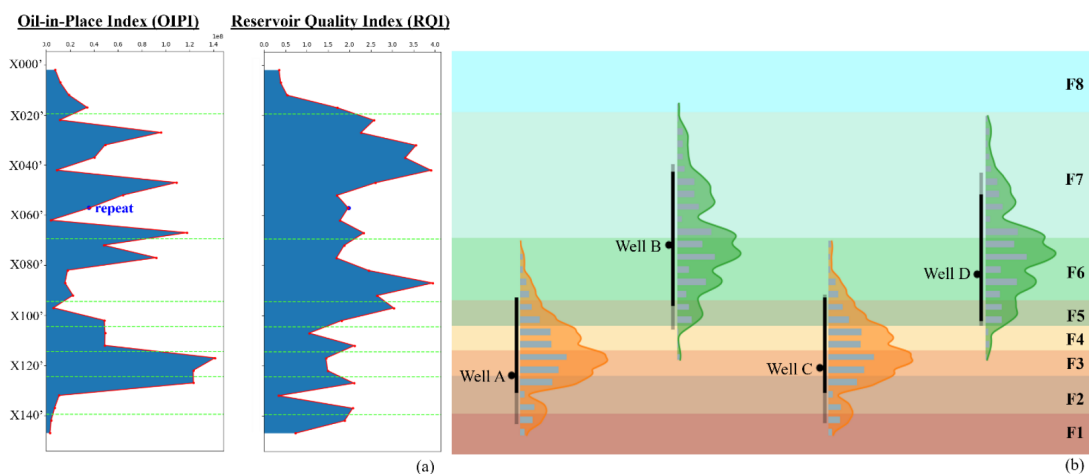


Figure 2—(a) Geochemically derived Reservoir Characterization Indices, Oil-in-Place Index and Reservoir Quality Index plotted with anonymized depths. b) FDFC plot of last produced oil sample showing the oil drainage from each of the 4 producing wells in the studied pad, samples were taken ~5-weeks into production.

To optimize full-field development, best lateral spacing for each draining interval must be identified. Another novel use for GCxGC geochemical fingerprint data is through the comparison of produced oils from neighboring wells which allows for an understanding of inter-well communication. An algorithm referred to as Similarity Index (SI) compares geochemical fingerprints of neighboring wells to understand lateral and/or vertical DRV overlap between adjacent wells. When Similarity Index data is partnered with production data

it can be determined what SI values correlate with optimal spacing. The Similarity Index tool is specifically outlined in [Ge et al. 2022](#).

This project utilizes 30 cores collected at a 5' sampling frequency that cover 145' TVD from formations F1 through F8. Oil was extracted from each rock endmember to build the baseline model. Produced oils were collected from 4 individual wells off the same pad, and in close proximity to the baseline, over the first ~30 – 36 days of production. Production allocation was calculated for each sample. Similarity Index Vertical and Lateral were calculated between produced oils of adjacent wells using samples that were collected on the same date for accurate representation of subsurface oil drainage.

When a pad of producing wells includes both Production Allocation (PA) and Similarity Index (SI), a visual tool known as DRV Mapping is provided. Figure 1c is a DRV map of 5 wells and demonstrates the drainage intensity around each producing well of a pad. The drainage intensity is constrained vertically by the PA result and laterally by the SI Lateral result. DRV Maps can also be normalized to production for enhanced visualization of drainage as shown in [Bachleda et al. 2024](#) and [Cisneros et al 2025](#).

Results and Discussion

Reservoir Characterization Indices and Production Allocation

Geochemically derived Reservoir Characterization Indices (RCI) from the nearby baseline well reveal the best mobile oil saturation (OIPI) and highest matrix permeability intervals (RQI) over the 145' analyzed interval. Shown in Figure 2a, the highest and most continuous OIPI is observed in F3 through uppermost F2 formation and corresponds to the location of the lower target. The second highest OIPI is observed intermittently throughout F7 into the top of F6. Best RQI is observed in the upper half of F7 and lower half of F6 into top F5. The upper landing zone is strategically located in the F6 formation to access high OIPI in the F7 and upper F6 formation as well as benefit from best matrix permeability as it is positioned between the highest RQI intervals.

Production allocation of all wells from ~5-weeks into production is visualized in the FDFC plot shown in Figure 2b. Wells landed in formation F6 drain up into F7, down into F4, and primarily from F7 and F6 formations with limited contribution from F8 or F3. Well B is landed ~10' shallower within formation F6 than Well D and by the ~5th week of production drains ~10% shallower, having ~45% F7 while Well D has ~35% F7; Well D has more contribution from F6, F5 and F4. Overall, by ~5-weeks into production formation F6 landed wells drain ~35 – 45% F7, ~40 – 50% F6, ~10 – 15% F5, and have minor F4 contribution.

Wells landed in formation F3 drain up into F6, down into F1, and primarily from F4 and F3. By ~5-weeks into production the wells see ~5-10% F6, ~10-15% F5, ~20-25% F4, ~30-35% F3, ~15-20% F2 and ~5% F1. Well C drains ~4% shallower than Well A, having slightly more F6, F5 and F4 contribution while Well A drains slightly more F3, F2 and F1. Wells landed in formations F6 and F3 drain primarily independent reservoir and see ~75 – 85% drainage difference between the two landing zones. All wells drain some from formations F6, F5 and F4 but when wells are spaced appropriately this staggered development pattern allows for optimal reservoir utilization.

Monitoring producing wells through time can provide key insights into reservoir properties driving early production or when notable inter-well communication occurs ([Bachleda et al. 2023](#), [Maxwell et al. 2024](#)). Wells B and C are centrally located in the development and see limited drainage change from the ~1st week to ~5th week of production. Wells A and D are the outer wells of the development and see ~7 – 9% drainage change over the first ~5-weeks of production. This limited drainage change over early production is one indication that the wells have limited SRV overlap with one another and are not in strong well communication.

Another way to visualize drainage change is through monitoring the drainage centroid movement through time. The drainage centroid is derived from drainage height weighted by its corresponding production contribution; it is calculated by multiplying depths and their relative PA percentage ([Ge et al. 2023](#)). The

drainage centroid is often observed near the landing zone, indicating the landing zone is a primary control on zonal contributions. However, at times the drainage centroid can have significant deviation from the landing zone which would indicate there are other major controls on the zonal contributions such as important reservoir properties, reservoir depletion, inter-well communication or unexpected SRV growth caused by unique pressure scenarios such as completion timing, or parent-child interactions (Kaufman et al. 2024).

Figure 3 visualizes drainage for Wells A and D through time in several ways. Figure 3a shows drainage of Well A from day 1 to day 36 of production, the drainage centroid is also shown through time. On the first day of production the drainage centroid is $\sim 5'$ above the landing zone, indicating landing zone is a key control on zonal contributions for wells landed in F3 formation. Over the first 36 days drainage shallows by a total of 9% as contribution from F6, F5, F4 and F3 increase through time, contributions from F2 and F1 correspondingly decrease. The drainage centroid also moves further above the landing zone which may be indicating that higher permeability intervals seen in F5 and F6 intervals are able to contribute more through time while the lower OIPI interval at the base of F2 into F1 is less able to sustain long-term production. Well D sees the drainage centroid $\sim 12'$ above the landing zone and has the opposite trend, seeing drainage deepen $\sim 7\%$ from the 5th day of production to the 33rd day of production. As drainage deepens contributions from F6, F5 and F4 increase through time, F7 contribution correspondingly reduces, and the drainage centroid moves closer to the landing zone. This subtle change may also be hinting a similar story as seen in Well A drainage change, that the higher permeability intervals in lower F6 and upper F5 are able to contribute more through time. Drainage Frac Heights (DFH) are also notably larger in F6 landed wells compared to F3 landed wells, the average P80 DFH (i.e., 80% of interval contributing to production) in F6 wells is $\sim 49 - 52'$ and $\sim 38'$ in F3 landed wells. This may indicate that having higher permeability reservoirs above and below the landing zone allows for larger SRV growth when compared to wells that do not have higher permeability reservoirs both above and below the landing zone. It has also been observed that wells landed in intervals with higher OIPI do not need as large of DFH to be successful producers (Ge et al. 2023). Wells landed in formation F3 see lower DFH but also have more prolific reserves in formations F3 – upper F2.

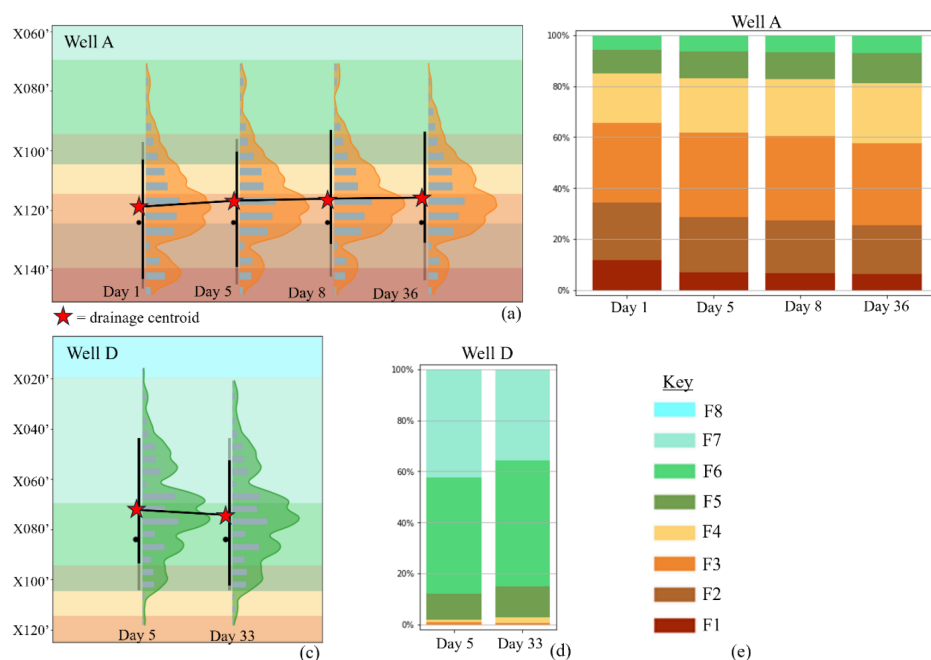


Figure 3—Drainage of Wells A and D through time. a) Well A time-lapse Production Allocation (PA) viewed in the Fitted Drainage Frac Curve (FDFC) plot with drainage centroid shown in the red star, highlighting end-member drainage change through time; b) Well A time-lapse PA in the stacked plot, highlighting zonal allocation change through time; c) Well D time-lapse PA viewed in the FDFC plot with drainage centroid shown in the red star; d) Well D time-lapse PA in the stacked plot; e) formation color key.

Similarity Index Findings and DRV Mapping

In an effort to understand overall reservoir utilization and potential Drainage Rock Volume (DRV) overlap between adjacent wells, Similarity Index (SI) has been calculated on same day samples for all 4 wells in the pad. SI Lateral (SIL) utilizes geochemical fingerprints that are more sensitive to lateral geochemical change to best understand amount of lateral frac-wing DRV overlap between wells landed in the same bench. An SIL of 10 is a laboratory repeat produced oil sample and indicates the exact same DRV, while the SIL approaches zero as the drainage becomes more different. SIL is shown in Figure 4a and 4b for the 1st and 5th week of production for both F6 landed wells and F3 landed wells. The F6 well pair sees SIL of 1.0 that increases to 2.0 while the F3 well pair sees an SIL of 2.1 increase to 2.9 over early production. These SIL values indicate moderate lateral DRV overlap that is increasing through time. While F3 wells see slightly higher SIL, there is still no indication these wells are spaced too closely. To understand when wells are spaced too closely SI data must be combined with production data (Ge et al. 2022, Bachleda et al 2024, Cisneros et al 2025). Permian Basin examples have shown that when SIL exceeds 6 wells are spaced too closely and competing for resources. Vaca Muerta examples have shown when SIL nears or exceeds 5 that wells are spaced too closely. A higher permeability reservoir as shown in this example may have a much lower SIL for optimal spacing.

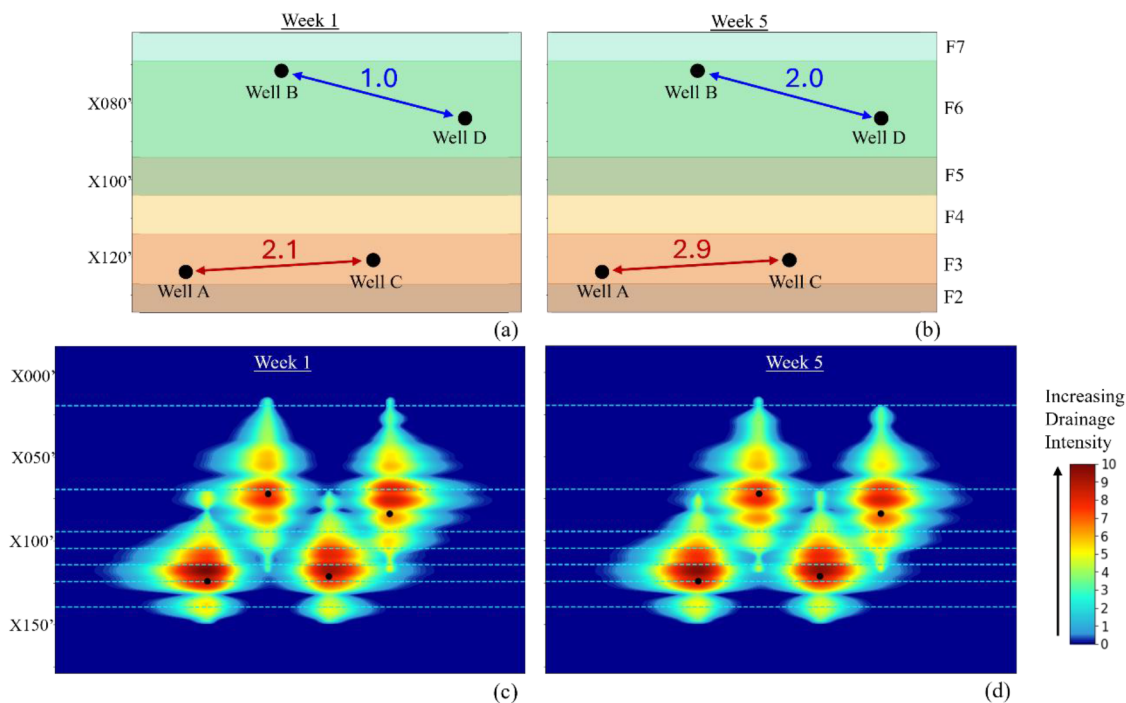


Figure 4—SIL (Similarity Index Lateral) and DRV (Drainage Rock Volume) mapping results for weeks 1 and 5 of production.
 a) SIL results of produced oil samples collected the 1st week of production; b) SIL results of produced oil samples collected the 5th week of production; c) DRV Mapping of PA and SI results from produced oil samples collected the 1st week of production; d) DRV Mapping of PA and SI results from produced oil samples collected the 5th week of production.

The SIL and PA data is further visualized in Figure 4c and 4d using the DRV mapping tool. Drainage intensity in all wells is highest near the landing zone, echoing that landing zone is a key driver to zonal contributions. The space between wells shows increased drainage intensity through time (week 1 to week 5) which reveals how the wells are draining to further extents laterally through time, leaving no by-passed pay behind. These DRV maps provide a visual to the quantitative PA and SIL data, all data indicating that these four wells are placed appropriately for excellent reservoir utilization and are not spaced too closely.

Conclusions

A comprehensive workflow is presented that applies high resolution geochemistry-based methods to accomplish reservoir characterization and monitoring. This workflow is utilized to provide quantitative production allocation from each interval in unconventional reservoirs at ~5 – 10' resolution, and to evaluate inter-well communication between neighboring wells. One example from a 4-well pad with target 2 separate landing zones is provided and results from the first ~5-weeks of production are reviewed. Key learnings reveal:

- The different landing zones in formations F6 and F3 drain primarily independent reservoir despite having less than ~100' of vertical targeting difference. F6 and F3 landed wells have ~75 – 85% drainage difference confirming the importance of developing the interval with 2 landing zones. F6 landed wells drain primarily from F7 and F6 formations, F3 landed wells drain primarily from F4 and F3 formations, but all wells drain some from formations F6, F5 and F4.
- Similarity Index (SI) is used to understand Drainage Rock Volume (DRV) overlap between adjacent wells and reveals the 4-wells are stacked and spaced appropriately, confirming cube-development of the 4-well pad allows for excellent reservoir utilization at current spacing.
- Wells B and D drilled in formation F6 with higher average geochemically derived Reservoir Quality Index (RQI), a matrix permeability indicator, see ~30% larger P80 Drainage Frac Heights (DFH) than wells A and C drilled in the highest OIPI interval with lower matrix permeability. This indicates having higher permeability rock both above and below the landing zone may allow for larger DFH. Wells landed in formation F3 see lower P80 DFH but also have more prolific reserves in formations F3 – upper F2.
- Geochemistry-based Reservoir Characterization Indices (RCI) aid in understanding what reservoir properties are contributing to production. Higher permeability reservoir in lower F6 and upper F5 are able to contribute more through time in both F6 and F3 landed wells.

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